

Meeting Summaries of March 10 and 12

Date: March 10 (with Jian) and March 12

Time1: 17:00 - 17:40 PM(PDT)

Time2: 18:00 - 18:40 PM(PDT)

Attendees1: Jian and Feng

Attendees2: Prof. Jan, Dr. Javed, Dr. Yue, Jian and Feng

Topic:

Summary of Discussion

- On March 10, Jian gave a presentation on the work of Benthic Assessment of Sediment via multivariate approach, which corresponds to Feng's current work. Specifically,
 1. Jian talked the reasons for using **multivariate approach** and picking the **16 out of 27 variables** in pollution assessment.
 2. Jian shared the ecological reasons and motivations why the **16 taxa** are good indicators for sediment contamination. Afterwards, Jian explained how their characters and behaviors support clustering to elucidate their spatial distribution in the environmental gradient space.
 3. The forward-stepwise regression was not used in Feng's work, so Jian stopped by that part. The hindsight 'relabeling' operation in the LDA work was not explained in the meeting, it was too long to remember why it was done. But this 'relabeling' operation will be replaced by a proper method after trying some new methods in the book - **Enumerative Ecology with R**, specifically, the chapter **4.12** - Multivariate Regression Trees (MRT) and **4.15** - Fuzzy Clustering for Noise Clustering.
- On March 12, the meeting was around the crucial deadlines of thesis, how the following communication should be around the specific writing and results to better use the time before and speed up the completion of the thesis. Some information and suggestions around the thesis and graduation defence and are:
 1. Dr. Javed informed that: **external expert** will be chosen by the committee and finalized by the research office. In May, it needs to start looking for the external expert and he or she will not be revealed to and approached by the student.
 2. Prof. Jan suggested Feng needs to go back to the **proposal** and underline each of the objectives outlined in that, make sure they will be 'solved' in the thesis chapters.
 3. Dr. Javed: analysis will **not** always favor the **expectation**, modification and explanation are inevitably needed, back and forth within and between analysis and writing. It needs room and preparation to handle these uncertainties (risks), instead of the idealistic schedule of weekly plans that follows a linear pace of making chapters and results, the worse case needs to be prepared and handled earlier.
 4. Prof. Jan: if some methods and their results are not working and helpful a lot, using **your supervisors** as good resources to decide what to do and there are always solutions.

5. Prof. Jan and Dr. Javed: the weekly writing work should be done on time (Monday), but it does **not** practically make sense to comment on every paragraph every week. It is worth to **briefly summarize** the written part and focus on the important and **uncertain parts**, so that it checks the weekly completion of writing and gives review and confirmation on the uncertain parts.
6. Prof. Jan: first attack the parts that you are **least certain** about. Do it **across the chapters** to find, sort and solve these most uncertain parts first, not in a linear way of finishing each single chapter one by one, which is idealistic and not practical.
7. From the contents of the thesis, the Simulation approach to apply **Power Analysis** is missing, it needs to be added following the quantile regression part, set as a new **Chapter 6**.
8. The chemical data of 1991 is under preparation and it might come later, the computations are safe to be done on the current complete data set.
9. The ideal defence date is in **July**, and the defence usually needs **one month** before the submission of final thesis draft.

Action Items

- Feng will be writing thesis and the first version of **completion** is first priority. The way to finish it needs **follow the suggestions** of "across chapters, the most uncertain parts first, then stay back to the easier work".
- The weekly meeting will be kept in this way, and before the meeting there worth to share the slides with summary of the work, which divides into **two major parts**:
 1. What regular content and work has been done (account for 20% in slides)
 2. What uncertain parts have been tried and how the results used in the thesis (account for 80% in the slides)

Follow-Up

Temporary meeting time: Mar 19, 18:00 - 18:40 PM(PDT)

Supportive materials

- A meeting recording with subtitles has been restored on Feng's local drive.